Hotel Booking Cancellation Predictor

Machine Learning Classification Project · IronHack Data Analytics Bootcamp

Gabriela Cascione

The Problem

28%

of bookings
are cancelled

Revenue Loss

Empty rooms can't be resold at short notice

Operational Chaos

Staffing, catering and room prep all wasted

Our Goal

Predict cancellations before they happen

Binary Classification: Will this booking be cancelled? (0 = No / 1 = Yes)

Dataset Selection

2 datasets evaluated — 1 selected

✅ **Hotel Booking Demand**

Kaggle — jessemostipak · 119,390 rows · 32 columns

CHOSEN — Clean, well-documented, rich features, clear binary target variable (is_canceled)

❌ **Hotel Reservations Dataset**

Kaggle — ahsan81 · Similar structure, less rows

Not selected — fewer features, less documentation

Data Cleaning

119,390	86,914	30
Raw Rows	After Cleaning	Final Columns

01 **Dropped agent & company**
Over 90% null values — not usable for modelling

02 **Dropped nulls in children & country**
Small number of affected rows — safe to remove entirely

03 **Removed 31,984 duplicate rows**
Identical rows would skew model training and evaluation

04 **Dropped data leakage columns**
reservation_status, reservation_status_date, assigned_room_type

Feature Engineering & Encoding

Country → Continent

170+ country codes grouped into 7 regions using pycountry_convert. Reduces noise, preserves geographic signal

Month → Number

arrival_date_month converted from string to 1–12. Enables tree models to learn seasonal patterns

Label Encoding (not OHE)

Categorical columns mapped to integers. Keeps feature count low and preserves interpretable feature importance scores

StandardScaler Normalisation

All numerical features scaled to zero mean and unit variance — essential for distance-based model

Model Comparison

9 classification models trained and evaluated · Sorted by F1 Score

Model	Accuracy	Precision	Recall	F1 Score
Gradient Boosting	**0.8105**	**0.7039**	**0.5455**	**0.6146**
Extra Trees	0.8119	0.7129	0.5378	0.6131
Random Forest	0.8155	0.7442	0.5091	0.6046
AdaBoost	0.7996	0.6754	0.5326	0.5955
XGB	0.8053	0.7254	0.4784	0.5766
Decision Tree	0.7718	0.6035	0.5139	0.5551
Bagging	0.7995	0.7221	0.4489	0.5536
KNN	0.7899	0.6946	0.4313	0.5322
Logistic Regression	0.7743	0.6963	0.3289	0.4468

Feature Selection

Feature importances from top model: Gradient Boosting — features with importance < 0.01 were dropped

Top 3 Predictors

lead_time
20.3% importance score

average_daily_rate
14.5% importance score

arrival_date_day_of_month
9.2% importance score

Dropped Features

importance < 0.01	
children	0.009
distribution_channel	0.007
prev_bookings_not_canceled	0.007
babies	0.002
is_repeated_guest	0.001
days_in_waiting_list	0.001

Before vs After Reduction

Model	Accuracy	F1
Original	0.8113	0.616
Reduced	0.8102	0.614

No improvement → kept full feature set

After dropping 6 low-importance features, model was retrained. Accuracy and F1 both dropped slightly — confirming the original full feature set was optimal.

Hyperparameter Tuning

RandomizedSearchCV · GradientBoostingClassifier · cv=5 · n_iter=10

RandomizedSearch tries 10 random combinations, balancing speed and coverage

n_estimators — [100, 250, 500]
Number of boosting stages

max_depth — [5, 10, 20]
Max depth of individual trees

max_leaf_nodes — [100, 250, 500]
Max leaf nodes per tree

⏱️ **Training time: ~109 minutes · Key reason for choosing RandomizedSearch**

Result

62.4%

F1 Score

Best params:

n_estimators: 250 | max_leaf_nodes: 100 | max_depth: 10

F1 improved by +0.01 vs original

Challenges & Next Steps

Challenges

Index misalignment after encoding

Train/test index mismatch caused NaN rows after encoding — fixed by resetting index inside the encoding function

Unseen values in test set

Country → Continent conversion had gaps — resolved by checking all unique values on the full dataset before splitting

Long training times

Several cells took significant time to run, limiting the number of iterations and experiments possible within the week

Next Steps

Address Class Imbalance

72/28 imbalance (not cancelled/cancelled) identified but not addressed due to time. Next: verify best method and apply

Deeper Hyperparameter Search

RandomizedSearchCV was used for speed. A full GridSearchCV on more combinations could improve results further

Try LightGBM / CatBoost

More powerful gradient boosting variants that were not explored this week

Hotel Booking
Cancellation Predictor

62,41% F1 Score · 9 models trained · Gradient Boosting wins

Gabriela Cascione
IronHack Data Analytics Bootcamp

Thank you — Questions welcome!